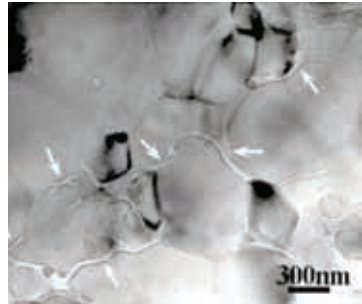


reinforcement surface area means that a relatively small amount of nanoscale reinforcement can have an observable effect on the macroscale properties of the composite. For example, adding carbon nanotubes improves the electrical and thermal conductivity.

They are generally divided into:

1. Ceramic-matrix nanocomposites:

This group of composites is mainly made of ceramic (a chemical compound from the group of oxides, nitrides, borides, silicides etc) and a metal as the second component, which are finely dispersed in each other in order to elicit the particular nanoscopic properties.

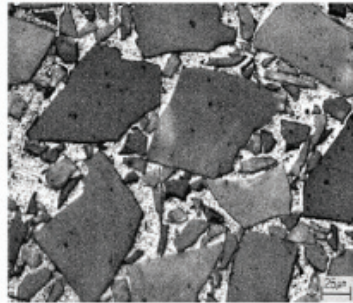


Ceramic-matrix nanocomposites

2. Metal-matrix nanocomposites:

This energy nanocomposite when combined with metal oxides and nano-scale aluminum powder, forms superthermite materials.

- 3. Polymer nanocomposites:** This group of composites is mainly made from a polymer or copolymer having dispersed nanoparticles. An example of a nanopolymer is silicon nanospheres which show quite different characteristics; their size is 40 – 100 nm and they are much harder than silicon, their hardness being between that of sapphire and diamond.



Metal matrix nanocomposite

Applications of Nanocomposites under Development:

- ▶ Producing batteries with greater power output.
- ▶ Speeding up the healing process for broken bones.
- ▶ Producing structural components with a high strength-to-weight ratio
- ▶ Using graphene to make composites with even higher strength-to-weight ratios.
- ▶ Making lightweight sensors
- ▶ Making flexible batteries.
- ▶ Making tumors easier to see and remove.